November 10, 2026

Hybrid finite element and neural network methods for parametric PDEs

Michel Duprez[1], Emmanuel Franck[2],
Frédérique Lecourtier[1] and Vanessa Lleras[3]

[1]Project-Team MIMESIS, Inria, Strasbourg
[2]Project-Team MACARON, Inria, Strasbourg
[3]IMAG, University of Montpellier, Montpellier

MACS seminar
("Modélisation, Analyse et Calcul Scientifique")

Context and motivation

Many-query contexts: real-time simulation, digital twins, optimization, uncertainty quantification

$\Rightarrow$ **Goal:** solve a parametric PDE accurately and quickly, for many parameters $\boldsymbol{\mu}$

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Our focus: the expressivity of the approximation spaces

Two hybridizations

How does the neural network enter the approximation space of a classical method?

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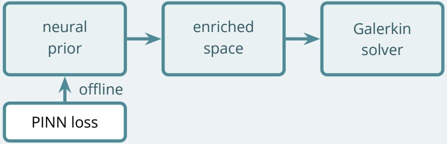
1st hybridization

Enriched FEM using PINNs


neural prior
enriched space
Galerkin solver
offline
PINN loss

Network: pretrained, frozen

Guarantees: a priori error estimates

Results: 1D, 2D, 3D, parametric

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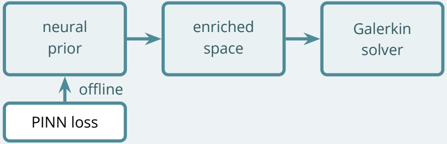
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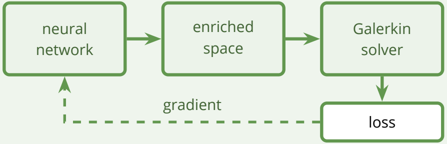
Learnable Galerkin methods


neural network
enriched space
Galerkin solver
gradient
loss

Network: learned through the solver

Guarantees: none (experimental)

Results: 1D, 3D, parametric

Parametric elliptic PDEs

Let $\Omega \subset \mathbb{R}^d$ be the spatial domain ($d = 1, 2, 3$) and $\mathcal{M} \subset \mathbb{R}^p$ the parameter space.
For one or several $\boldsymbol{\mu} \in \mathcal{M}$, find $u(\,\cdot\,; \boldsymbol{\mu}) : \Omega \to \mathbb{R}$ such that

$$\begin{cases} \mathcal{L}\big(u\,; \boldsymbol{x}, \boldsymbol{\mu}\big) = f(\boldsymbol{x}, \boldsymbol{\mu}), & (\boldsymbol{x}, \boldsymbol{\mu}) \in \Omega \times \mathcal{M}, \\ \text{Dirichlet, Neumann or Robin BC}, & (\boldsymbol{x}, \boldsymbol{\mu}) \in \Gamma \times \mathcal{M}, \end{cases} \tag{$\mathcal{P}$}$$

with $\Gamma := \partial\Omega$ the boundary of Ω, f the source term and $\mathcal{L}$ the linear differential operator defined by

$$\mathcal{L}(\cdot\,; \boldsymbol{x}, \boldsymbol{\mu}) : u \mapsto R(\boldsymbol{x}, \boldsymbol{\mu})\, u + \boldsymbol{C}(\boldsymbol{x}, \boldsymbol{\mu}) \cdot \nabla u - \frac{1}{\text{Pe}} \nabla \cdot \big(D(\boldsymbol{x}, \boldsymbol{\mu}) \nabla u\big).$$

where R is the reaction, $\boldsymbol{C}$ the convection, D the (symmetric positive definite) diffusion matrix and Pe the Péclet number.

Numerical methods

Numerical methods for parametric PDEs*

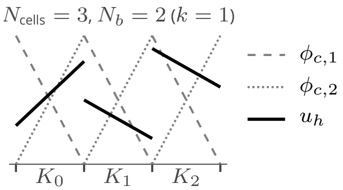
* presented with homogeneous Dirichlet boundary conditions

Discontinuous Galerkin (DG) method[1]

For a fixed $\boldsymbol{\mu} \in \mathcal{M}$, let $\mathcal{T}_h$ be a mesh of Ω of cells $(K_c)_{0 \leq c < N_{\text{cells}}}$.

Let V_h^{disc} be a discontinuous space of piecewise polynomials of degree k on the mesh $\mathcal{T}_h$. Its basis functions $(\phi_{c,i})_{c,i}$, with $\phi_{c,i}$ supported on a single cell K_c, decompose the discrete solution:

$$u_h(\boldsymbol{x}, \boldsymbol{\mu}) = \sum_{c,i} \beta_{c,i}(\boldsymbol{\mu})\, \phi_{c,i}(\boldsymbol{x}).$$



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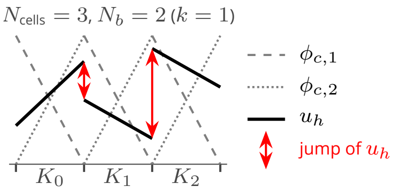

> **Coupling the cells:** ambiguous traces on the interfaces $\Rightarrow$ numerical flux (it defines the method)
>
> Diffusion-dominated regimes: flux for the diffusion term only, here SIPG (Symmetric Interior Penalty Galerkin)

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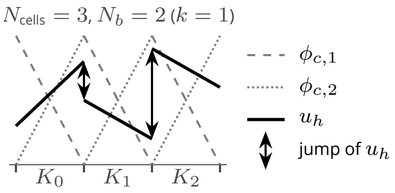

Variational problem (SIPG):

Find $u_h \in V_h^{\text{disc}}$ such that, $\forall v_h \in V_h^{\text{disc}}$, $a_h^{\text{disc}}(u_h, v_h; \boldsymbol{\mu}) = l(v_h; \boldsymbol{\mu}) = \int_\Omega f\, v_h$, with

$$a_h^{\text{disc}}(u_h, v_h; \boldsymbol{\mu}) = \boxed{\int_\Omega \left(R\, u_h v_h + \boldsymbol{C} \cdot \nabla_h u_h\, v_h + \frac{1}{\text{Pe}} D \nabla_h u_h \cdot \nabla_h v_h \right)} \ \text{cell terms}$$

penalty

$$\text{consistency}\ \boxed{- \frac{1}{\text{Pe}} \int_{\mathcal{F}_h} \left([\![u_h]\!] \cdot \{D \nabla_h v_h\} + \{D \nabla_h u_h\} \cdot [\![v_h]\!] \right)} \ \boxed{+ \frac{1}{\text{Pe}} \int_{\mathcal{F}_h} \kappa\, [\![u_h]\!] \cdot [\![v_h]\!]}$$

where $\mathcal{F}_h$ is the set of faces, ∇_h the broken gradient, $[\![\cdot]\!]$ the jump, $\{\cdot\}$ the average, κ the penalty parameter.

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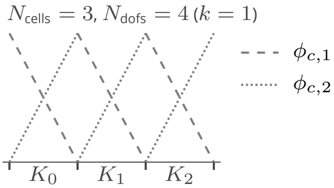
[1] [Arnold et al., 2002]

Continuous Galerkin (CG) method[1]

Same mesh $\mathcal{T}_h$ and fixed $\boldsymbol{\mu}$ as DG.

Let $V_h := V_h^{\text{disc}} \cap C^0(\Omega)$ be a continuous space of piecewise polynomials of degree k on $\mathcal{T}_h$. Its basis functions $(\phi_I)_I$, with ϕ_I supported on the cells sharing the dof I, decompose the discrete solution:

$$u_h(\boldsymbol{x}, \boldsymbol{\mu}) = \sum_{I=1}^{N_{\text{dofs}}} \beta_I(\boldsymbol{\mu})\, \phi_I(\boldsymbol{x}).$$



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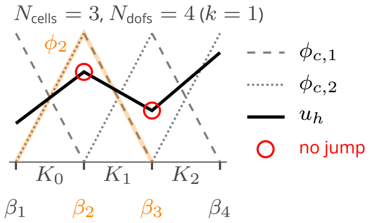

Coupling the cells: shared dofs via the connectivity table ⇒ no jumps, no numerical flux

$$\text{conn} : (c, i) \mapsto I, \quad \beta_I(\boldsymbol{\mu}) = \beta_{c,i}(\boldsymbol{\mu}), \quad \phi_I = \textstyle\sum_{\text{conn}(c,i)=I} \phi_{c,i}$$

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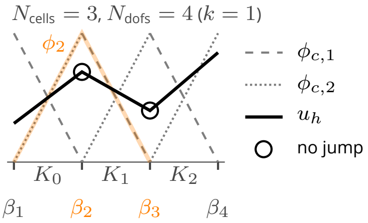

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cell terms

with $V_h^0 \subset V_h$ vanishing on Γ and no interface terms, no penalty.

---

[1] [Ern and Guermond, 2004]

Galerkin scheme

Linear system (in both methods):

$$\underbrace{A_h(\boldsymbol{\mu})}_{\text{matrix}}\ \underbrace{\boldsymbol{\beta}_h(\boldsymbol{\mu})}_{\text{dofs}} = \underbrace{\boldsymbol{b}_h(\boldsymbol{\mu})}_{\text{right-hand side}}$$

	DG	CG
coupling	weak, via numerical fluxes	strong, in the space
$\boldsymbol{\beta}_h(\boldsymbol{\mu})$	$\big(\boldsymbol{\beta}_c(\boldsymbol{\mu})\big)_c\ ,\ \ \boldsymbol{\beta}_c(\boldsymbol{\mu}) = (\beta_{c,i}(\boldsymbol{\mu}))_i$	$\big(\beta_I(\boldsymbol{\mu})\big)_I\ ,\ \ I = \text{conn}(c,i)$

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$A_h(\boldsymbol{\mu})$	$A_{(c,i),(c',j)}(\boldsymbol{\mu}) = a_h^{\text{disc}}(\phi_{c',j}, \phi_{c,i}; \boldsymbol{\mu})$	$A_{IJ}(\boldsymbol{\mu}) = a(\phi_J, \phi_I; \boldsymbol{\mu})$
$\boldsymbol{b}_h(\boldsymbol{\mu})$	$b_{(c,i)}(\boldsymbol{\mu}) = l(\phi_{c,i}; \boldsymbol{\mu})$	$b_I(\boldsymbol{\mu}) = l(\phi_I; \boldsymbol{\mu})$

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$\boldsymbol{b}_h(\boldsymbol{\mu})$	$b_{(c,i)}(\boldsymbol{\mu}) = l(\phi_{c,i}; \boldsymbol{\mu})$	$b_I(\boldsymbol{\mu}) = l(\phi_I; \boldsymbol{\mu})$
structure	larger system, one block per cell size: $N_{\text{cells}} \cdot N_b$	smaller system, dofs shared between cells size: N_{dofs}

Physics-informed neural networks (PINNs)[1]

Let $u_\theta(\boldsymbol{x}, \boldsymbol{\mu})$ be a neural network with weights θ and inputs $(\boldsymbol{x}, \boldsymbol{\mu}) \in \Omega \times \mathcal{M}$, approximating the solution u.

Training: Find the weights $\theta^\star$ such that

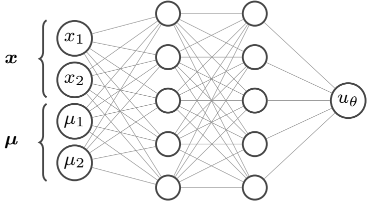
$$\theta^\star = \underset{\theta}{\arg\min} \left(\omega_r \, \mathcal{J}_r(\theta) + \omega_b \, \mathcal{J}_b(\theta)\right), \qquad (\mathcal{P}_\theta)$$

with ω_r and ω_b the weights of the two loss terms, and

residual loss
$$\boxed{\mathcal{J}_r(\theta) = \int_{\mathcal{M}} \int_{\Omega} \left| \mathcal{L}\left(u_\theta(\boldsymbol{x}, \boldsymbol{\mu}); \boldsymbol{x}, \boldsymbol{\mu}\right) - f(\boldsymbol{x}, \boldsymbol{\mu}) \right|^2 \mathrm{d}\boldsymbol{x} \, \mathrm{d}\boldsymbol{\mu},}$$

boundary loss
$$\boxed{\mathcal{J}_b(\theta) = \int_{\mathcal{M}} \int_{\Gamma} \left| u_\theta(\boldsymbol{x}, \boldsymbol{\mu}) \right|^2 \mathrm{d}\boldsymbol{x} \, \mathrm{d}\boldsymbol{\mu}.}$$

Monte-Carlo method:
Discretize the cost functions by random process.



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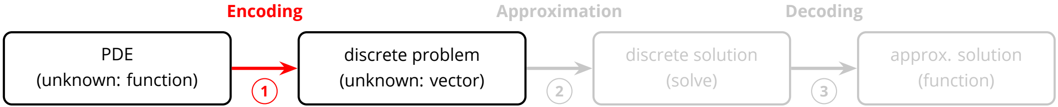
[1] [Raissi, Perdikaris, and Karniadakis, 2019]

Link between Galerkin and PINNs

Numerical method (three steps):


Encoding
Approximation
Decoding
PDE
(unknown: function)
discrete problem
(unknown: vector)
discrete solution
(solve)
approx. solution
(function)
①
②
③

	Galerkin (DG/CG)	PINN
Space	vector space (V_h^{disc}/V_h)	manifold of the u_θ
① Unknowns	dofs $\boldsymbol{\beta}_h(\boldsymbol{\mu})$	weights θ
Integrals	quadrature on cells	Monte Carlo on collocation points

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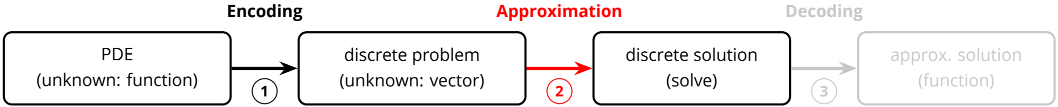
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②	Parameters	fixed $\boldsymbol{\mu}$, one solve per $\boldsymbol{\mu}$	$\boldsymbol{\mu} \in \mathcal{M}$ as input, one training
	Resolution	linear system $A_h(\boldsymbol{\mu})\boldsymbol{\beta}_h(\boldsymbol{\mu}) = \boldsymbol{b}_h(\boldsymbol{\mu})$	optimization on θ

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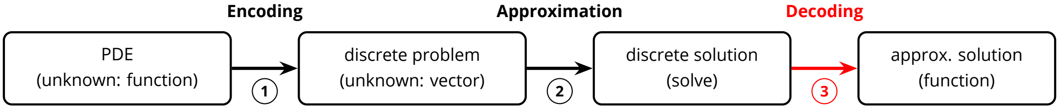

Encoding **Approximation** **Decoding**

PDE (unknown: function) → ① → discrete problem (unknown: vector) → ② → discrete solution (solve) → ③ → approx. solution (function)

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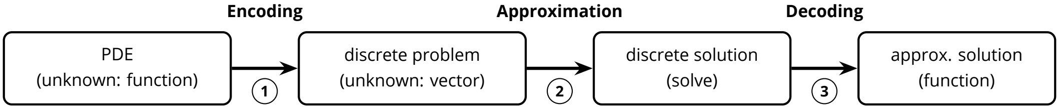
③	Solution	$u_h(\boldsymbol{x}, \boldsymbol{\mu})$, linear in $\boldsymbol{\beta}_h$	$u_\theta(\boldsymbol{x}, \boldsymbol{\mu})$, non-linear in θ

Link between Galerkin and PINNs

Numerical method (three steps): **Same philosophy, different approximation space**



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1st hybridization

Enriched finite element method using PINNs

Method

Pipeline of the hybrid method

TODO

Schéma bleu : PINN paramétrique hors ligne, prior figé, FE enrichi en ligne

Additive approach

TODO

Prior, espace enrichi V_h^+, problème variationnel

Multiplicative approach

TODO

Relèvement M, espace enrichi $V_h^\times$, problème variationnel

Convergence analysis

TODO

FEM, gain additif, gain multiplicatif, limite $M \to \infty$

Numerical results

1D Poisson
2D Poisson
3D Poisson

Numerical results

1D Poisson
2D Poisson
3D Poisson

1D Poisson: additive vs multiplicative

TODO

Courbes d’erreur : FEM, additif, multiplicatif ($M = 3$, $M = 100$)

Numerical results

1D Poisson
2D Poisson
3D Poisson

2D Poisson on a square

TODO

Courbes d’erreur, N_{dofs} à erreur égale, gain moyen

Numerical results

1D Poisson
2D Poisson
3D Poisson

3D Poisson: online phase

TODO

Erreur en h et en temps en ligne, speed-up

3D Poisson: parametric framework

TODO

Coût total hors ligne + en ligne, nombre de paramètres de rentabilité

2nd hybridization

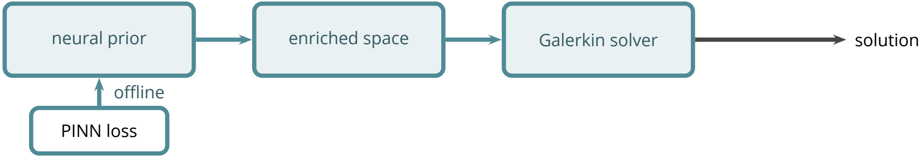
Learnable Galerkin methods

Introduction

From frozen prior to learned basis

1st hybridization: neural prior trained offline, then frozen: a priori error estimates.

Limit: the gain is bounded by the quality of the prior, trained apart from the scheme.



neural prior → enriched space → Galerkin solver → solution

PINN loss → (offline) neural prior

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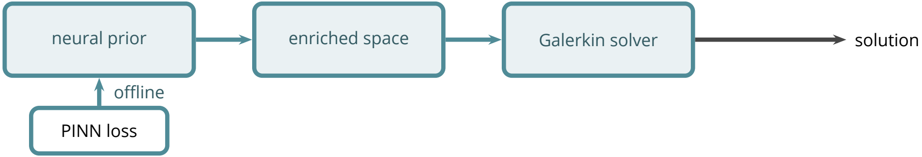

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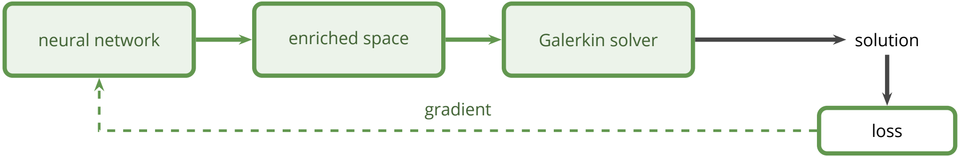
2nd hybridization: learn the basis through the solver, adapted to the solution of the scheme.

Advantage: basis adapted to the scheme, not bounded by a frozen prior.

Limit: no a priori guarantees, training cost.



neural network → enriched space → Galerkin solver → solution

solution → loss

loss → (gradient) → neural network

Automatic differentiation and ScimBa

Need: gradient of the solver w.r.t. the N_θ weights, without deriving it by hand.

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Three ways to compute the gradient:

	exact	**use**	**cost per gradient**
symbolic	✓	closed form only	huge expressions
finite differences	×	any program	N_θ solves
automatic diff.[1]	✓	any program	a few solves

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[1] [Baydin et al., 2018]

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ScimBa: the whole pipeline is a JAX program[2]:

network → basis → assembly → solve → loss $\mathcal{J}$

One backward pass gives $\partial \mathcal{J} / \partial \theta$ through all the steps.

See ScimBa

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[1] [Baydin et al., 2018]

[2] [Bradbury et al., 2018]

Non-parametric enrichment (PINN loss)*

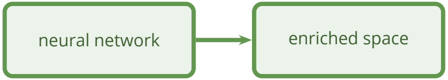
* $\boldsymbol{\mu}$ is fixed and omitted from the notation: basis, dofs and solution are valid for this $\boldsymbol{\mu}$ only

Neural-enriched Galerkin scheme

neural network

Neural network: Construct a single neural network $v_\theta(\boldsymbol{x})$ on Ω (with N_θ weights), shared by all cells.

Neural-enriched Galerkin scheme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enriched space

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Enriched basis: Construct a multiplier $m_\theta(\boldsymbol{x})$ from $v_\theta(\boldsymbol{x})$ and define the enriched basis functions[1] as:

$$\phi_{c,i}^{\theta}(\boldsymbol{x}) = m_\theta(\boldsymbol{x})\, \phi_{c,i}(\boldsymbol{x}).$$

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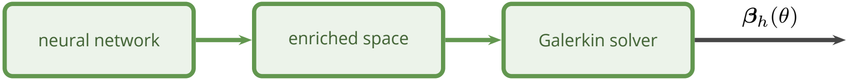
[1] can be adapted to CG using the connectivity table

Neural-enriched Galerkin scheme



neural network → enriched space → Galerkin solver → $\boldsymbol{\beta}_h(\theta)$

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Galerkin scheme ($\boldsymbol{\mu}$ fixed): Solve the linear system:

$$\boldsymbol{R}_h\big(\boldsymbol{\beta}_h(\theta);\theta\big) = A_h(\theta)\,\boldsymbol{\beta}_h(\theta) - \boldsymbol{b}_h(\theta) = \boldsymbol{0},$$

with $$A_{(c,i),(c',j)}(\theta) = a_h^{\text{disc}}(\phi^\theta_{c',j}, \phi^\theta_{c,i}), \qquad b_{(c,i)}(\theta) = l(\phi^\theta_{c,i}).$$

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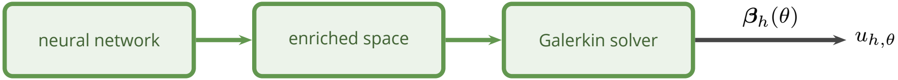
Neural-enriched Galerkin scheme



neural network

enriched space

Galerkin solver

$\boldsymbol{\beta}_h(\theta)$

$u_{h,\theta}$

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Solution: Reconstruct the solution in the enriched space as:

$$u_{h,\theta}(\boldsymbol{x}) = \sum_{c,i} \beta_{c,i}(\theta)\, \phi^\theta_{c,i}(\boldsymbol{x}).$$

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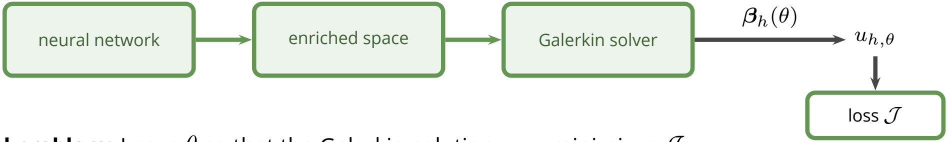
[1] can be adapted to CG using the connectivity table

Learning θ: differentiating through the solver


neural network

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Galerkin solver

$\boldsymbol{\beta}_h(\theta)$

$u_{h,\theta}$

loss $\mathcal{J}$

Bi-level problem: Learn θ so that the Galerkin solution $u_{h,\theta}$ minimizes $\mathcal{J}$:

$$\theta^\star = \arg\min_\theta \ \mathcal{J}(\boldsymbol{\beta}_h(\theta);\theta) \quad \text{s.t.} \quad \boldsymbol{R}_h(\boldsymbol{\beta}_h(\theta);\theta) = \mathbf{0}$$

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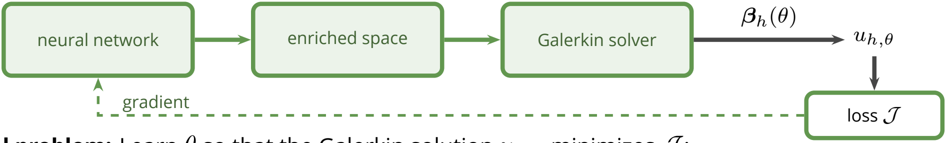
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gradient

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Gradient: Differentiate through the solver (chain rule):

$$\frac{d\mathcal{J}}{d\theta} = \underbrace{\left.\frac{\partial \mathcal{J}}{\partial \theta}\right|_{\boldsymbol{\beta}_h}}_{\text{direct}} + \underbrace{\frac{\partial \mathcal{J}}{\partial \boldsymbol{\beta}_h} \cdot \frac{d\boldsymbol{\beta}_h}{d\theta}}_{\text{through the dofs}}$$

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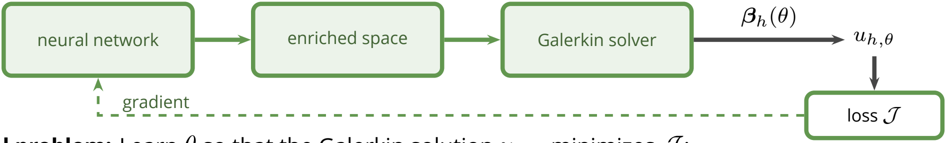

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Adjoint: With a gradient-based optimizer, avoid $d\boldsymbol{\beta}_h/d\theta$ (only one linear solve, whatever N_θ):

$$A_h(\theta)^T \boldsymbol{\lambda}_h = \left(\frac{\partial \mathcal{J}}{\partial \boldsymbol{\beta}_h}\right)^T \Rightarrow \frac{d\mathcal{J}}{d\theta} = \left.\frac{\partial \mathcal{J}}{\partial \theta}\right|_{\boldsymbol{\beta}_h} - \boldsymbol{\lambda}_h^T \frac{\partial \boldsymbol{R}_h}{\partial \theta}(\boldsymbol{\beta}_h;\theta)$$

1D test case and PINN loss

Test case: 1D advection-diffusion with a boundary layer, $\Omega = (0,1)$, Pe $= 20$, $f = 1$:

$$\begin{cases} -\frac{1}{\text{Pe}}\,\partial_{xx}u + \boldsymbol{C}\,\partial_x u = f & \text{in } \Omega, \\ u = 0 & \text{on } \Gamma, \end{cases} \qquad \boldsymbol{C}(x,\boldsymbol{\mu}) = \mu_1 + \mu_2 x$$

with $\boldsymbol{\mu} = (\mu_1, \mu_2) = (0.8, 0.8)$ fixed.

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PINN loss: Minimize residual and boundary terms ($\omega_r = 1$, $\omega_b = 10$):

$$\mathcal{J}(\theta) = \omega_r \underbrace{\int_\Omega \left| -\frac{1}{\text{Pe}}\,\partial_{xx} u_{h,\theta} + (\mu_1 + \mu_2 x)\,\partial_x u_{h,\theta} - f \right|^2 \mathrm{d}\boldsymbol{x}}_{\mathcal{J}_r \text{ (Monte Carlo, redrawn each epoch)}} + \omega_b \underbrace{\int_\Gamma \left| u_{h,\theta} \right|^2 \mathrm{d}\boldsymbol{x}}_{\mathcal{J}_b}$$

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Validity: Learned for this configuration only:

new $\boldsymbol{\mu}$, k or N_{cells} $\Rightarrow$ new training.

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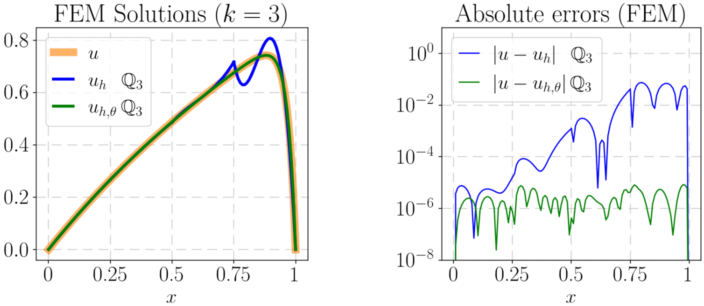
> **Evaluation:** Relative L^2 error w.r.t. a reference FEM solution (128 cells, $\mathbb{Q}_4$), never used in the training.

Learned basis: affine multiplier

Setting: CG, $N_{\text{cells}} = 4$ cells, $\mathbb{Q}_3$ ($N_{\text{dofs}} = 13$), affine multiplier $m_\theta = 2 + 0.1\, v_\theta$.
Training: MLP v_θ (tanh, 2×16, $N_\theta = 321$), ENG optimizer[1], 300 epochs, 6000 collocation points.



Solutions: Classical u_h oscillates in the boundary layer, $u_{h,\theta}$ matches the reference ($\sim 10^{-5}$ against $7 \cdot 10^{-2}$).

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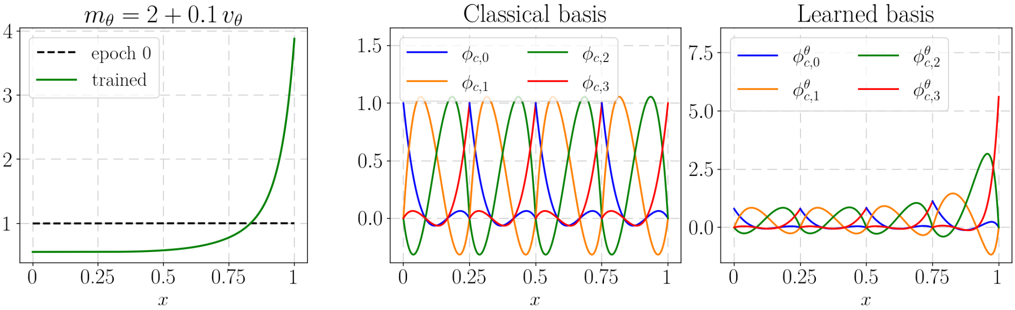

Solutions: Classical u_h oscillates in the boundary layer, $u_{h,\theta}$ matches the reference ($\sim 10^{-5}$ against $7 \cdot 10^{-2}$).
Basis: The multiplier grows in the boundary layer: the last basis functions are amplified.

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[1] [Müller and Zeinhofer, 2023]

Choice of the multiplier

Fix $N_{\text{dofs}} = 25$ and $N_\theta \in \{134, \dots, 151\}$, vary the levers: (N_{cells}, k) and the MLP shape (width, depth).

20 configurations $\times$ 20 seeds $= 400$ trainings per multiplier (100 epochs, 5000 collocation points).

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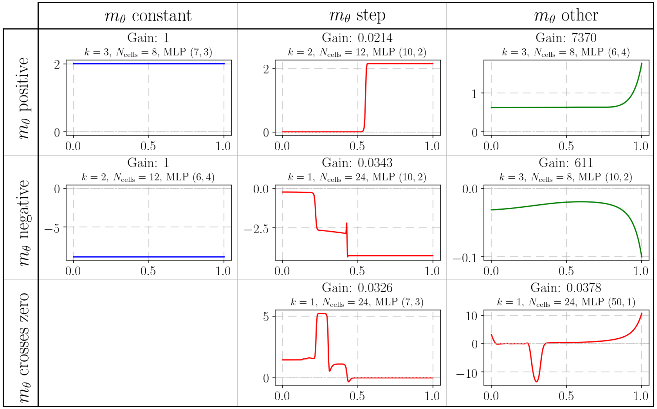
Affine multiplier:

$$m_\theta = 2 + 0.1\, v_\theta$$

Learned multiplier $m_\theta(x)$, one example per sign and shape



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Affine multiplier:

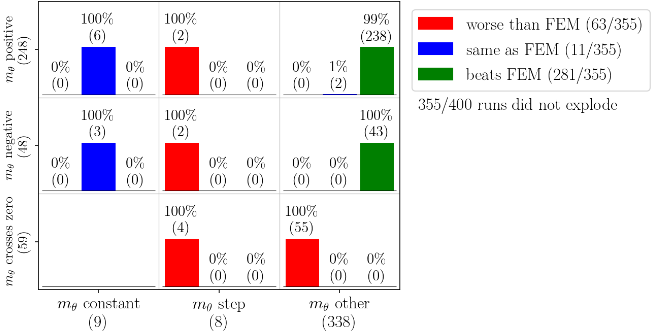
$$m_\theta = 2 + 0.1\, v_\theta$$

Diagnosis:

Some trainings explode (loss $=$ NaN).

Steps and sign changes never beat FEM.

Non-exploded runs by sign and shape of $m_\theta(x)$



355/400 runs did not explode

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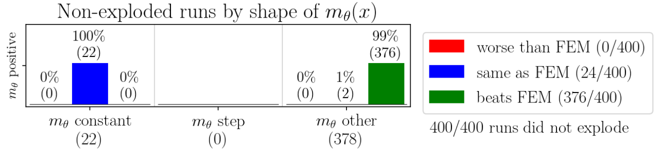


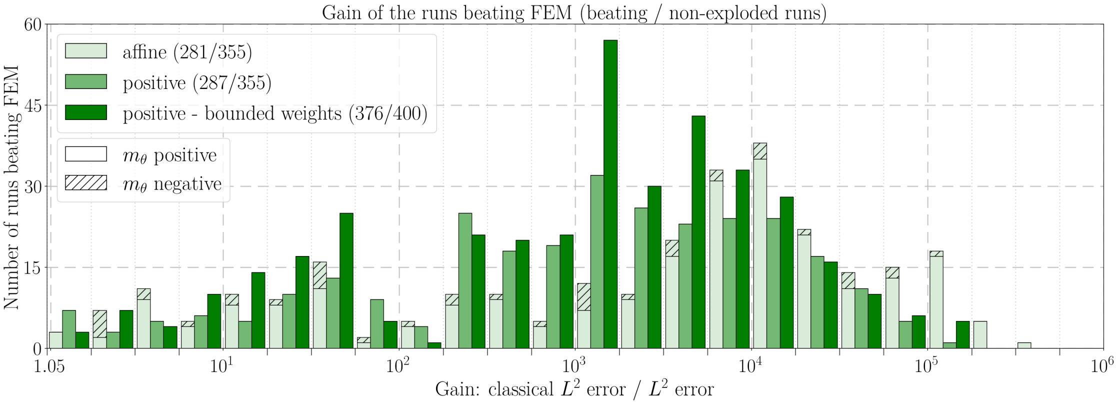
Fix: Positive multiplier

$$m_\theta = \exp(2 \tanh v_\theta)$$

with bounded weights:

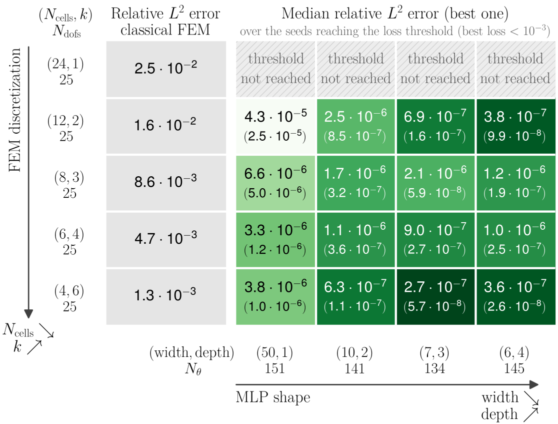
No sign change, no step.

Results at fixed budget



Gain: $376/400$ beat the FEM, none explodes; the affine multiplier keeps the highest gains.

Results at fixed budget


FEM discretization
$N_{\text{cells}} \searrow$
$k \nearrow$
MLP shape
width $\searrow$
depth $\nearrow$

Median relative L^2 error (best one)
over the seeds reaching the loss threshold (best loss $< 10^{-3}$)

(N_{cells}, k) N_{dofs}	Relative L^2 error classical FEM	$(50, 1)$ 151	$(10, 2)$ 141	$(7, 3)$ 134	$(6, 4)$ 145
$(24, 1)$ 25	$2.5 \cdot 10^{-2}$	threshold not reached	threshold not reached	threshold not reached	threshold not reached
$(12, 2)$ 25	$1.6 \cdot 10^{-2}$	$4.3 \cdot 10^{-5}$ $(2.5 \cdot 10^{-5})$	$2.5 \cdot 10^{-6}$ $(8.5 \cdot 10^{-7})$	$6.9 \cdot 10^{-7}$ $(1.6 \cdot 10^{-7})$	$3.8 \cdot 10^{-7}$ $(9.9 \cdot 10^{-8})$
$(8, 3)$ 25	$8.6 \cdot 10^{-3}$	$6.6 \cdot 10^{-6}$ $(5.0 \cdot 10^{-6})$	$1.7 \cdot 10^{-6}$ $(3.2 \cdot 10^{-7})$	$2.1 \cdot 10^{-6}$ $(5.9 \cdot 10^{-8})$	$1.2 \cdot 10^{-6}$ $(1.9 \cdot 10^{-7})$
$(6, 4)$ 25	$4.7 \cdot 10^{-3}$	$3.3 \cdot 10^{-6}$ $(1.2 \cdot 10^{-6})$	$1.1 \cdot 10^{-6}$ $(3.6 \cdot 10^{-7})$	$9.0 \cdot 10^{-7}$ $(2.7 \cdot 10^{-7})$	$1.0 \cdot 10^{-6}$ $(2.5 \cdot 10^{-7})$
$(4, 6)$ 25	$1.3 \cdot 10^{-3}$	$3.8 \cdot 10^{-6}$ $(1.0 \cdot 10^{-6})$	$6.3 \cdot 10^{-7}$ $(1.1 \cdot 10^{-7})$	$2.7 \cdot 10^{-7}$ $(5.7 \cdot 10^{-8})$	$3.6 \cdot 10^{-7}$ $(2.6 \cdot 10^{-8})$

(width, depth) N_θ

Gain: $376/400$ beat the FEM, none explodes; the affine multiplier keeps the highest gains.

Levers: degree and depth matter: $k = 1$ only a modest gain over FEM, $(50, 1)$ is the worst network.

3D test case

Test case: 3D advection-diffusion with a boundary layer, $\Omega = (0,1)^3$, Pe $= 20$:

$$\begin{cases} -\dfrac{1}{\mathsf{Pe}}\,\Delta u + \boldsymbol{C} \cdot \nabla u = f & \text{in } \Omega, \\ u = 0 & \text{on } \Gamma, \end{cases} \qquad \boldsymbol{C}(\boldsymbol{x}, \boldsymbol{\mu}) = (\mu_1 + \mu_2\, \boldsymbol{x} \cdot \boldsymbol{e}_{\text{diag}})\, \boldsymbol{e}_{\text{diag}},$$

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Exact solution: manufactured, vanishing on the 6 faces (f deduced from it):

$$u(\boldsymbol{x}) = \prod_{i=1}^{3} x_i \left(1 - \exp\left(-\frac{\text{Pe}\,(\mu_1 + \mu_2)}{\sqrt{3}}(1 - x_i)\right)\right)$$

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Evaluation: Relative L^2 error w.r.t. the manufactured solution u, never used in the training.

3D: learned multiplier

Setting: CG, $N_{\text{cells}} = 4 \times 4 \times 4$ cells, $\mathbb{Q}_2$ ($N_{\text{dofs}} = 729$), positive multiplier $m_\theta = \exp\left(6 \tanh v_\theta\right)$.
Training: MLP v_θ (tanh, 3×10, $N_\theta = 271$), ENG optimizer, 70 epochs, 6000 collocation points, 3 seeds.

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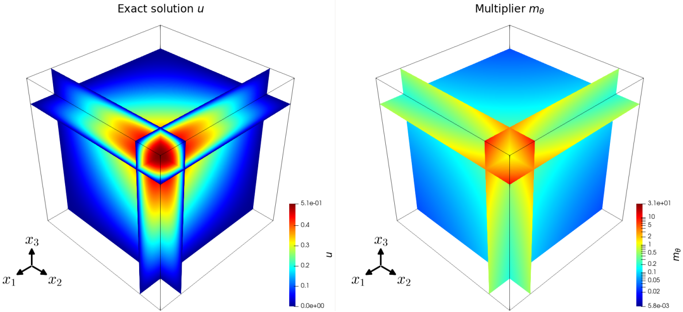

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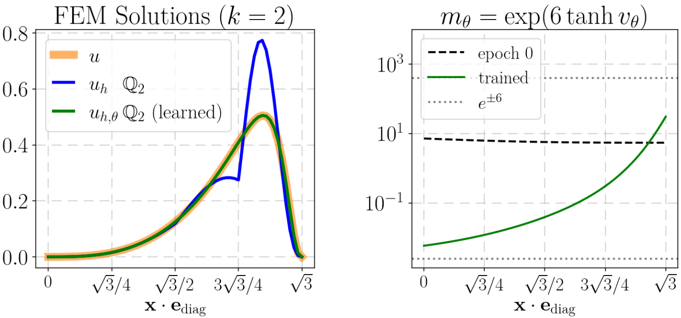

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Solution: classical u_h overshoots, $u_{h,\theta}$ matches u.

3D: gain and cost

	time (s)									
	w/ comp.		**w/o comp.**		N_{dofs}			**error**		
seed	**learn.**	**class.**	**learn.**	**class.**	**learn.**	**class.**	**ratio**	**learn.**	**class.**	**ratio**
1	59	51	10.1	49.5	729	$5.31 \cdot 10^5$	729	$5.68 \cdot 10^{-5}$	$3.53 \cdot 10^{-4}$	6.2
2	41	37	6.8	26.6	729	$3.89 \cdot 10^5$	534	$8.13 \cdot 10^{-5}$	$4.82 \cdot 10^{-4}$	5.9
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Equal time: error $\times 6$ lower for seeds 1, 2, equal for seed 3.

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Dofs: 534 to 729 times fewer for the learned basis (without counting the $N_\theta = 271$ weights).

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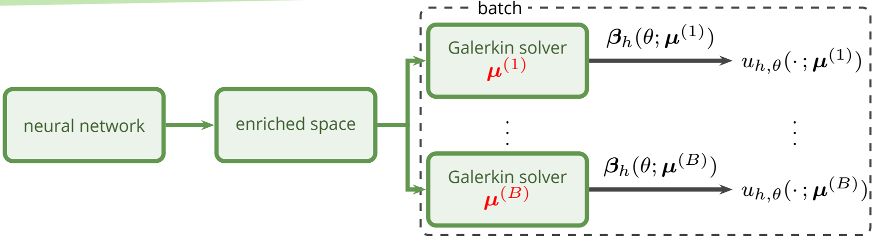
Cost: learned time dominated by compilation: 7 to 10 s without it, against 27 to 50 s for the FEM.

Parametric enrichment (neural operator)*

* $\boldsymbol{\mu}$ is no longer fixed: one network trained once, for all $\boldsymbol{\mu} \in \mathcal{M}$

Neural operator: principle


batch
neural network
enriched space
Galerkin solver
$\boldsymbol{\mu}^{(1)}$
Galerkin solver
$\boldsymbol{\mu}^{(B)}$
$\boldsymbol{\beta}_h(\theta; \boldsymbol{\mu}^{(1)})$
$u_{h,\theta}(\cdot\,; \boldsymbol{\mu}^{(1)})$
$\boldsymbol{\beta}_h(\theta; \boldsymbol{\mu}^{(B)})$
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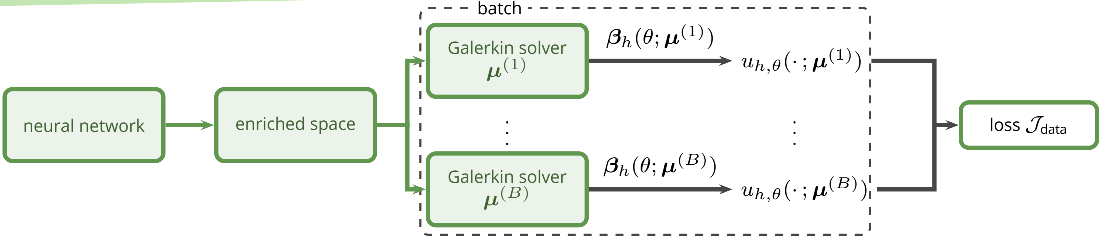
Parameter sampling: $\mathcal{D}_{\text{train}} \subset \mathcal{M}$ chosen once, a random batch of $\boldsymbol{\mu}$ drawn at each epoch.

Parametric scheme: One network, one Galerkin solve per $\boldsymbol{\mu}$ of the batch: $A_h(\theta; \boldsymbol{\mu})\, \boldsymbol{\beta}_h(\theta; \boldsymbol{\mu}) = \boldsymbol{b}_h(\theta; \boldsymbol{\mu})$.

Neural operator: principle


batch
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Galerkin solver
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Galerkin solver
$\boldsymbol{\mu}^{(B)}$
$\boldsymbol{\beta}_h(\theta; \boldsymbol{\mu}^{(B)})$
$u_{h,\theta}(\cdot\,; \boldsymbol{\mu}^{(B)})$
loss $\mathcal{J}_{\text{data}}$

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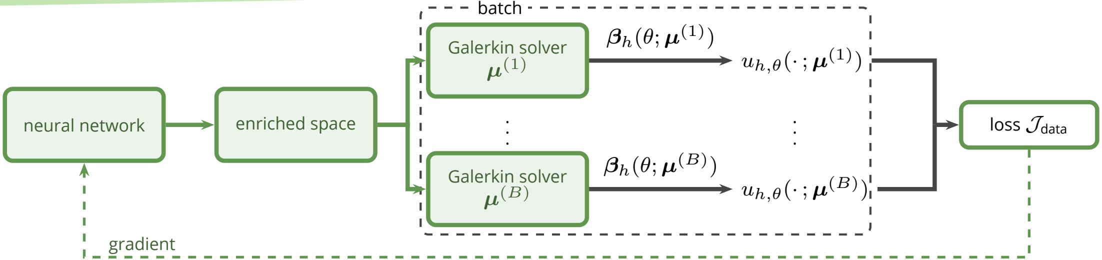
Supervised loss: Fit reference solutions on the batch, at N_x points per $\boldsymbol{\mu}$:

$$\mathcal{J}_{\text{data}}(\theta) = \frac{1}{B\, N_x} \sum_{i=1}^{B} \sum_{j=1}^{N_x} \left| u_{h,\theta}\big(\boldsymbol{x}^{(j)}; \boldsymbol{\mu}^{(i)}\big) - u_{\text{ref}}\big(\boldsymbol{x}^{(j)}; \boldsymbol{\mu}^{(i)}\big) \right|^2 .$$

Neural operator: principle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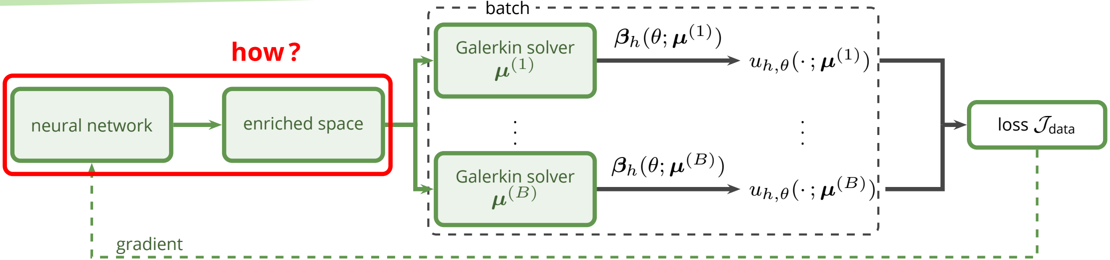
Neural operator: principle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how ?
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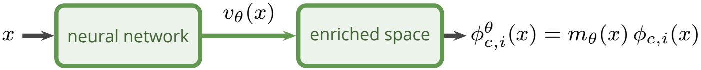
Constructing the enriched space (three ways)


x
neural network
$v_\theta(x)$
enriched space
$\phi^\theta_{c,i}(x) = m_\theta(x)\,\phi_{c,i}(x)$

Shared basis: same enriched space for all $\boldsymbol{\mu}$.

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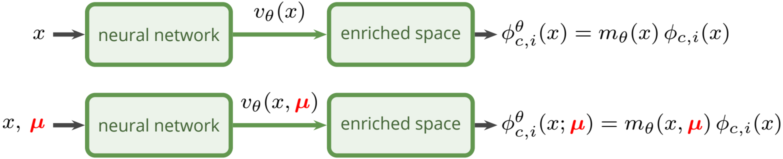
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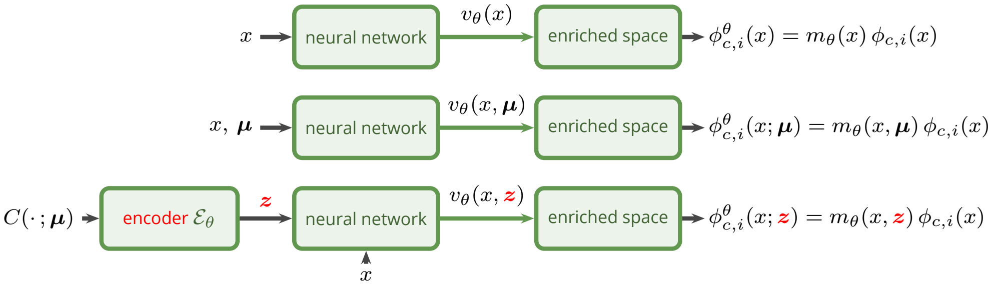
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$C(\cdot\,; \boldsymbol{\mu})$
encoder $\mathcal{E}_\theta$
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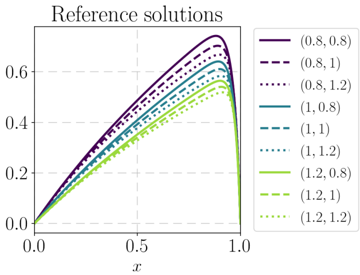
Field-dependent basis: an encoder summarizes the transport field in a latent state $\boldsymbol{z}$.

1D linear transport

Test case: Same 1D case as the PINN, with $\boldsymbol{C}(x, \boldsymbol{\mu}) = \mu_1 + \mu_2 x$, $\boldsymbol{\mu} \in [0.8, 1.2]^2$: the layer stays at $x = 1$.



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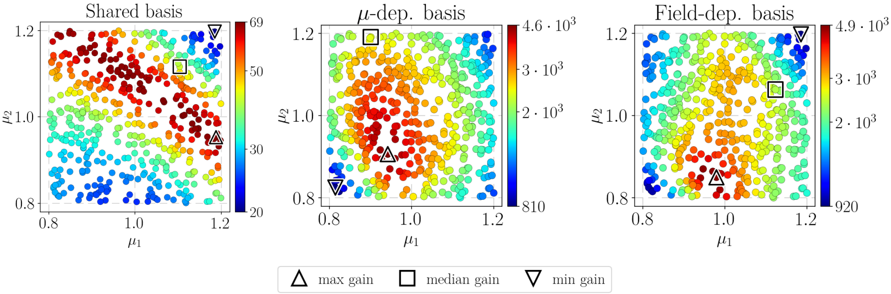
Setup: DG, $N_{\text{cells}} = 8$, $k = 2$, 32 training $\boldsymbol{\mu}$, batch of 16, $N_x = 101$ points per $\boldsymbol{\mu}$.



Gain (on 500 unseen $\boldsymbol{\mu}$):
shared 20–69 ($N_\theta = 321$), μ-dep. 811–$4.6 \cdot 10^3$ ($N_\theta = 353$), field-dep. 917–$4.9 \cdot 10^3$ ($N_\theta = 1381$).

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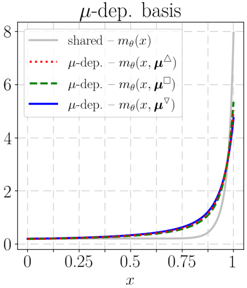

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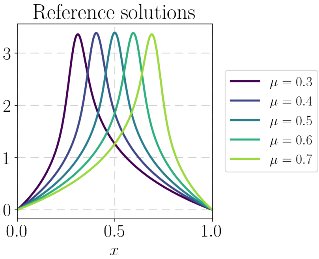
Multipliers: shared, rises only in the last cells; $\boldsymbol{\mu}$-dep., rises over more cells.

1D turning point

Test case: Same 1D equation as the PINN, $C(x, \mu) = \mu - x$, $\mu \in [0.3, 0.7]$, Pe $= 1000$: layer moves with μ.



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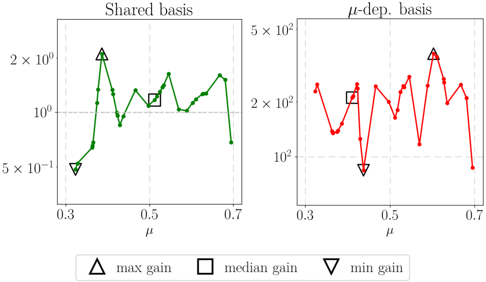

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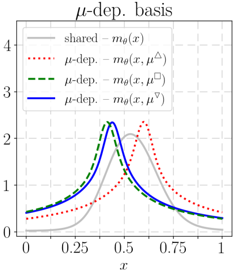

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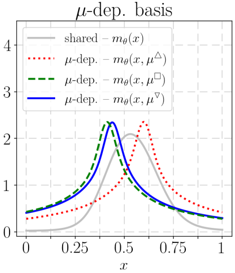

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- Convergence orders preserved, gain set by the quality of the prior (its derivatives)
- Additive, $\mathbb{P}_1$: meshes 5 to 28 times coarser for the same accuracy
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2nd hybridization: learnable Galerkin bases

- Differentiable solver: train the basis through the solver
- 1D and 3D: far fewer dofs for the same accuracy (1D: error 10^2 to 10^5 times lower at fixed N_{dofs})
- Parametric (1D, operator): unseen parameters, median gains from ≈ 9 to ≈ 7000

Limits and perspectives

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> **Link between the two:** freeze the learned multiplier $m_\theta \Rightarrow$ multiplicative approach of the 1st hybridization, a priori estimate (Céa-type).

Appendix – 1st hybridization

Enriched finite element method using PINNs

Additional details

General Dirichlet boundary conditions

TODO

CL de Dirichlet générales : additif, multiplicatif (PINN standard / CL exactes)

Multiplicative gain constants

TODO

Constantes de gain H^1 et L^2, limite $M \to \infty$

Extra results

Anisotropic 2D
Annulus 2D

Extra results

- Anisotropic 2D
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Anisotropic 2D: problem considered

TODO

Problème, PINN

Anisotropic 2D: online phase

TODO

Courbes d’erreur, gains

Anisotropic 2D: solutions and errors

TODO

Solutions et erreurs

Extra results

Anisotropic 2D

Annulus 2D

Poisson on an annulus: problem considered

TODO

Problème (CL mixtes), PINN

Poisson on an annulus: online phase

TODO

Courbes d'erreur, gains

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TODO

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Appendix – 2nd hybridization

Learnable Galerkin methods

Additional details

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TODO

Détails de la méthode (hors présentation principale)

Additional numerical results

Additional numerical results

TODO

Résultats numériques supplémentaires